

Data Structure and Algorithms

CSIT — 3rd Semester — 2082 — Answer Sheet
Subject Code: CSC211

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Distinguish between data type and abstract data type. List any three characteristics of algorithm. Evaluate the postfix expression $33 + 2 * 40$ -, using stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define height, depth and level of the tree. Find the minimum spanning tree of the following weighted graph using Kruskal and Prims algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**3. Define queue as ADT and discuss about primitive operations in queue. Trace the output generated by following recursive function for the statement TOH(4,1,3).
TOH(n, s, d){if (n == 1){printf(" move the top disk from rod %d to rod %d",s,d)}else{TOH (n- 1, s, 6- s- d)printf(" move the top disk from rod %d to rod %d",s,d)TOH (n- 1, 6- s- d, d)}}**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Why do we need algorithms for searching? Trace the binary search operation for the key = 6, in the set {3, 6, 8, 12, 67, 89, 90, 100}

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. How do you implement single linked list using stack? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Trace the merge sort algorithm for the input {8, 7, -4, 19, 3, 45, 12}.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain about any three hash collision resolution techniques.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Describe the process of deleting the node at start, end and specified position of singly linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Write the algorithm of Quick sort with its efficiency analysis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Why do we always prefer the worst case for algorithm analysis? Describe the insertion and deletion operation in BST.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define overflow and underflow. Discuss the advantages and limitations of circular queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What are the advantages of circular queue over linear queue? Sort the array {4, 7, 3, 2, 1} using insertion sort.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.